27. LVM Intro and /etc/fstab

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Phase: Phase 8 — Packages, disks, storage
Estimated time: 1–2 days
Slug: `27-lvm-fstab`

Learning objectives

- Explain PV/VG/LV mentally
- Read fstab entries
- Add mounts safely with backups
- Use UUIDs over volatile device names

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

LVM idea

Physical volumes → volume group → logical volumes (flexible resizing).

fstab

```
UUID=...  /data  ext4  defaults  0  2
```

Always backup fstab before edits. Wrong fstab can fail boot — have rescue plan (VM snapshot).

Common mistakes

- Using /dev/sdX names that reorder on reboot
- Editing fstab without reboot test in VM
- Shrinking filesystems without backups

Exercises

Work in `exercises/27-lvm-fstab/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/27-lvm-fstab/` until you have an honest failing attempt.

Self-check

1. Why UUID in fstab?

2. PV vs LV?
3. How recover bad fstab?

Further reading

- https://man7.org/linux/man-pages/man5/fstab.5.html